

FUNCTIONS AND GRAPHS, (SKETCHING QUADRATIC, CUBIC AND RECIPROCAL

Name: _____

Date: _____

LO: I can recognise and sketch quadratic, cubic and reciprocal graphs from their equations.

Types of graphs

A quadratic graph ($y = ax^2 + bx + c$) forms a U-shape (parabola). A cubic graph ($y = ax^3$) has an S-shape. A reciprocal graph ($y = \frac{a}{x}$) has two separate curves.

Key features: Look for the y-intercept, turning points, and asymptotes to sketch each graph accurately.

Graph shapes

●	$y = x^2$ is a U-shaped parabola	minimum at origin
●	$y = x^3$ passes through the origin with S-shape	point of inflection
●	$y = \frac{1}{x}$ has two branches in Q1 and Q3	asymptotes at axes
●	$y = x^2$ is a straight line	it is a curve

Key idea

The highest power of x tells you the shape: x^2 = parabola, x^3 = cubic curve, $\frac{1}{x}$ = reciprocal (hyperbola).

$$y = ax^2 + bx + c$$

a determines direction, c is the y-intercept

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - quadratic graph

Sketch $y = x^2 - 4$

Shape: U-shaped parabola (positive x^2)

y-intercept: when $x = 0$, $y = -4$

x-intercepts: $x^2 = 4$, so $x = 2$ and $x = -2$

U-shape crossing x-axis at (-2, 0) and (2, 0), vertex at (0, -4)

The -4 shifts the parabola down by 4 units.

Example 2 - negative quadratic

Sketch $y = -x^2 + 9$

Shape: inverted U (negative x^2)

y-intercept: (0, 9)

x-intercepts: $x^2 = 9$, so $x = \pm 3$

Inverted U with maximum at (0, 9), crossing at (-3, 0) and (3, 0)

Negative coefficient of x^2 flips the parabola.

Example 3 - cubic graph

Sketch $y = x^3 - x$

Factorise: $y = x(x^2 - 1) = x(x - 1)(x + 1)$

x-intercepts: $x = -1, 0, 1$

Shape: S-curve (positive x^3) through these three roots

S-curve crossing x-axis at -1, 0, and 1

Factorising helps find where the cubic crosses the x-axis.

Example 4 - reciprocal graph

Sketch $y = \frac{3}{x}$

Asymptotes at $x = 0$ and $y = 0$

When $x = 1$, $y = 3$; when $x = 3$, $y = 1$

When $x = -1$, $y = -3$; curves in Q1 and Q3

Two branches: one in Q1, one in Q3, never touching the axes

The graph gets closer to the axes but never reaches them.

YOUR TURN – PRACTICE (deliberate practice)

1. Identifying graph types

State whether each equation gives a quadratic, cubic, or reciprocal graph.

- a) $y = 2x^2 + 1$
 b) $y = x^3 - 3$
 c) $y = \frac{5}{x}$
 d) $y = -x^2 + 4x$

2. Key features of quadratics

Find the y-intercept and direction (U or inverted U).

- a) $y = x^2 + 3$
 b) $y = -x^2 + 5$
 c) $y = 2x^2 - 8$
 d) $y = -3x^2$

3. Finding x-intercepts

Find where each graph crosses the x-axis (set $y = 0$).

- a) $y = x^2 - 9$
 b) $y = x^2 - 5x + 6$
 c) $y = x^3 - 4x$
 d) $y = x^2 + 2x - 8$

4. Sketching from features

List the key features needed to sketch each graph.

- a) $y = x^2 - 2x - 3$
 b) $y = x^3$
 c) $y = \frac{2}{x}$
 d) $y = -x^3$

5. Mixed graph problems

Answer each question about graphs.

- a) How many turning points does $y = x^3 - 3x$ have?
 b) What is the vertex of $y = (x - 2)^2 + c$?
 c) Does $y = \frac{1}{x}$ have any lines of symmetry?
 d) What happens to $y = \frac{1}{x}$ as x gets very large?
 e) Vertex of $y = x^2 - 6x + 8$?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $y = x^2 + 1$ never crosses the x-axis ☐

Correct answer: _____

Reason: _____

b) $y = \frac{1}{x}$ passes through the origin ☐

Correct answer: _____

Reason: _____

c) A negative coefficient of x^2 flips the parabola upside down ☐

Correct answer: _____

Reason: _____

d) Cubic graphs always have three x-intercepts ☐

Correct answer: _____

Reason: _____

e) The y-intercept is found by setting $x = 0$ ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) The height of a ball thrown upwards is modelled by $h = -5t^2 + 20t$ where h is height in metres and t is time in seconds. Sketch the graph. What is the maximum height? When does the ball land?

Expression: _____

Simplified: _____

b) A graph passes through (1, 6), (2, 3), and (3, 2). Show that this could be a reciprocal graph of the form $y = \frac{k}{x}$. Find k and predict the y -value when $x = 6$.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
 Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

